

Campylobacter fetus Infectious Disease

— Comprehensive Disease

Characteristics Report

Target disease: Campylobacter fetus Infectious Disease **MONDO ID:** MONDO:0040728
Category: Infectious disease (zoonotic, bacterial) **Report type:** Disease knowledge-base entry, structured per the 15-section research template

Summary

Campylobacter fetus infectious disease is a **zoonotic, foodborne bacterial infection** caused by the Gram-negative, microaerophilic, spiral bacterium *Campylobacter fetus*. Unlike the enteric ("thermophilic") campylobacters *C. jejuni* and *C. coli*, which chiefly cause self-limited diarrhea, *C. fetus* characteristically produces **invasive, extra-intestinal disease** — most importantly bacteremia with a striking predilection for the **vascular endothelium** (mycotic aneurysm, infective endocarditis, endovascular graft/prosthesis infection), together with meningitis, cellulitis, septic arthritis, spontaneous bacterial peritonitis, and transplacental infection causing abortion, stillbirth, and neonatal sepsis/meningitis. The disease concentrates in **elderly and immunocompromised adults**, with reported 30-day mortality of roughly **12–15%** and, in perinatal transmission, fetal/neonatal mortality as high as **80%**.

The central virulence mechanism is a **paracrystalline surface (S-) layer** encoded by a family of *sapA* homologs. The S-layer confers **complement/serum resistance**, enabling the organism to survive in the bloodstream, and it undergoes **high-frequency antigenic variation** by DNA inversion/reciprocal recombination, allowing immune evasion and relapse. This S-layer/complement-resistance axis, intersecting with **host immune competence**, governs disease severity: convergent in-vitro, mouse-model, and clinical evidence supports a single coherent causal model in which S-layer-mediated serum resistance is the upstream determinant of bacteremia, endovascular/CNS/placental seeding, and antigenic-variation-driven relapse, all amplified when host immunity is impaired.

Management relies on **aminoglycosides (gentamicin)**, **carbapenems (meropenem/imipenem)**, or **ampicillin**, because *C. fetus* is **intrinsically resistant to cephalosporins** and increasingly acquires **fluoroquinolone resistance** (via *gyrA/parC* mutations). Timely, appropriate antibiotic therapy is independently associated with survival. There is **no licensed human vaccine**; human prevention rests on food safety and hygiene around livestock and reptiles and on protecting vulnerable hosts. The species comprises three reservoir-defined subspecies — **subsp. fetus** (cattle/sheep; the main cause of human systemic disease), **subsp. venerealis** (bovine genital campylobacteriosis), and the emerging reptile-associated **subsp. testudinum**. This is an **acquired infectious disease with no heritable genetic basis**; the "genetic" content of this template therefore pertains to *bacterial* virulence and resistance genes rather than human germline variants.

1. Disease Information

Overview. *Campylobacter fetus* infectious disease denotes human (and animal) infection by the bacterium *C. fetus*. In humans it manifests predominantly as **systemic/invasive infection** — bacteremia with frequent secondary endovascular, neurological, musculoskeletal, soft-tissue, and perinatal localizations — rather than as the gastroenteritis typical of *C. jejuni/C. coli*. It is a rare but serious pathogen that "mainly affect[s] immunocompromised patients" ([PMID: 41241280](#)).

Key identifiers. - **Mondo:** MONDO:0040728 (*Campylobacter fetus* infectious disease) - **MeSH:** *Campylobacter* Infections (D002169); organism *Campylobacter fetus* (NCBI Taxonomy txid **196**) - **ICD-10:** A04.8 (Other specified bacterial intestinal infections) / A28.8 (other zoonotic bacterial diseases) depending on presentation; specific bacteremia coded per site (e.g., I77.x for mycotic aneurysm complications) - **ICD-11:** 1A0Z / 1G40-range "bacterial infection of unspecified site" plus site-specific codes - **OMIM / Orphanet:** Not applicable — this is an acquired infection, not a Mendelian or rare *genetic* disorder, so no OMIM phenotype number or Orphanet rare-disease code applies.

Synonyms / alternative names. *Campylobacteriosis* due to *C. fetus*; *Vibrio fetus* infection (historical name); *C. fetus* bacteremia/septicemia; "campylobacteremia" (in the context of *Campylobacter* bloodstream infection). In cattle: bovine genital campylobacteriosis (BGC)/bovine venereal campylobacteriosis (subsp. *venerealis*) and sporadic/epizootic bovine-ovine abortion (subsp. *fetus*).

Information source. The evidence base is **aggregated disease-level** literature — retrospective clinical cohorts, national bacteremia surveillance studies, case series/reviews, microbiological/genomic studies, and animal models — rather than individual EHR-derived patient records.

2. Etiology

Primary cause — infectious. The disease is caused by infection with *Campylobacter fetus*, a Gram-negative, oxidase-positive, microaerophilic, S-shaped/curved rod. There is **no genetic (human germline) etiology**; the causal factor is exposure to the bacterium, typically via the fecal–oral/foodborne route from animal reservoirs and subsequent translocation from the gut across the intestinal epithelium into the bloodstream (see §6).

Risk factors (host/environmental). The dominant, repeatedly documented risk factors are **advanced age and immunocompromise/comorbidity**: - Elderly (median ages 68–78 across cohorts) (PMID: 34849656; PMID: 17999095). - Immunodepression (43.4% of one national cohort), hematologic malignancy (25.9%), solid tumor (23%), diabetes (22.3%) (PMID: 34849656). - **Asplenia, rituximab** maintenance therapy, and **occupational exposure** (e.g., abattoir work) (PMID: 29984777). - HIV infection (associated with relapsing, quinolone-resistant disease) (PMID: 9534967). - Male sex predominance (e.g., 78% male in a meningitis review) (PMID: 41241280). - Dietary/animal exposures: unpasteurized dairy, undercooked meat, contact with cattle/sheep, and — for subsp. *testudinum* — contact with pet reptiles (turtles/tortoises) (PMID: 40484837; PMID: 31928704).

Genetic risk factors. No established human susceptibility loci. Host innate-immune competence (complement, LPS-responsiveness) modulates outcome — demonstrated in mice where LPS-responsive C3H/HeN strains suffered higher mortality than LPS-hyporesponsive C3H/HeJ (PMID: 2318963) — but no human polymorphisms are validated.

Protective factors. Intact immunity and spleen function; timely, appropriate antibiotic therapy is independently protective against death (OR 0.47, 95% CI 0.24–0.93) (PMID: 34849656). No genetic protective variants are described.

Gene–environment interactions. The operative interaction is **pathogen-gene × host–environment**: the bacterial S-layer (serum resistance) interacts with the host's immune status to determine whether gut translocation progresses to sustained bacteremia and metastatic seeding (§6, §11).

3. Phenotypes

C. fetus infection is clinically heterogeneous; presentations reflect the site of metastatic seeding after bacteremia. Onset is typically **adult/geriatric**, **acute-to-subacute**, and severity is **moderate-to-severe** (frequently life-threatening).

Phenotype	Type	Characteristics / frequency	Suggested HPO
Fever / febrile illness	Symptom/sign	Very common presenting feature; acute	HP:0001945 (Fever)
Bacteremia / sepsis	Lab + clinical	Hallmark; ~64% of analyzed invasive cases (21/33) had documented bacteremia (PMID: 37877803)	HP:0031864 (Bacteremia); HP:0100806 (Sepsis)
Mycotic aneurysm / endovascular infection	Clinical sign	Vascular tropism: 5/7 secondary localizations vascular, 3 mycotic aneurysm (PMID: 37877803); mycotic aortic aneurysm 24% (PMID: 17999095)	HP:0031649 (Mycotic aneurysm), HP:0004942 (Aortic aneurysm)
Infective endocarditis	Clinical sign	12 of 80 secondary-localization patients (PMID: 34849656)	HP:0100584 (Endocarditis)
Cellulitis	Physical manifestation	19% in two cohorts (PMID: 18699745 ; PMID: 17999095)	HP:0100658 (Cellulitis)
Meningitis	Clinical sign	Relapse 22%, mortality 5% in a 37-case review; median age 49, 78% male (PMID: 41241280)	HP:0001287 (Meningitis)
Septic arthritis / osteoarticular	Clinical sign	24 of 80 secondary-localization patients (PMID: 34849656)	HP:0012385 (Arthritis)
Spontaneous bacterial peritonitis / ascites	Clinical sign	9 of 80 secondary-localization patients (PMID: 34849656 ; PMID: 29853499)	HP:0030151 (Peritonitis)
Abortion / stillbirth / perinatal sepsis	Reproductive	18/20 pregnancies ended prematurely (13–32 wk); fetal/neonatal mortality 80% (PMID: 3523697)	HP:0100687 (Spontaneous abortion)
Diarrhea (less common)	Symptom	Possible but not the dominant presentation	HP:0002014 (Diarrhea)

Quality-of-life impact. Invasive disease carries substantial acute morbidity (ICU-level sepsis, vascular surgery for mycotic aneurysm, prolonged IV antibiotics) and, in survivors of endovascular or CNS disease, potential long-term functional impairment; relapsing infection (notably in HIV/immunocompromised hosts and meningitis) prolongs disease burden. Formal EQ-5D/SF-36 data specific to *C. fetus* are **not available**.

4. Genetic / Molecular Information

Not applicable as human genetics. *C. fetus* disease has **no causal human genes, pathogenic germline variants, modifier genes, or chromosomal abnormalities**. The relevant molecular determinants are **bacterial**:

- ***sapA* family (surface-array protein / S-layer proteins).** Multiple *sapA* homologs encode full-length S-layer proteins (98/127/149 kDa) attached serospecifically to type A or B lipopolysaccharide ([PMID: 7885229](#)). This is the principal virulence gene family (see §6).
- ***sapCDEF* — type I secretion system** exporting the S-layer proteins without an N-terminal signal sequence ([PMID: 9851986](#)).
- **Virulence/invasion genes:** *ciaB* (invasion), cytolethal distending toxin *cdtA/cdtB/cdtC*, and a **Type IV secretion system (T4SS)** plus *fic*-domain genes carried on genomic islands/plasmids ([PMID: 40484837](#); [PMID: 27049518](#)).
- **Resistance determinants:** *gyrA* QRDR mutations (e.g., Asp→Tyr; T86I) and *parC* mutations conferring fluoroquinolone resistance; *tet(O)* for tetracycline resistance; intrinsic cephalosporin resistance ([PMID: 9534967](#); [PMID: 22771419](#); [PMID: 40484837](#)).

"Epigenetic"/genomic plasticity (bacterial). Antigenic variation of the S-layer is achieved by **inversion of a promoter-containing invertible DNA element flanked by *sapA* homologs**, and by reciprocal DNA recombination among *sapA* copies — a programmed genomic-rearrangement mechanism rather than host epigenetics ([PMID: 7885229](#); [PMID: 9851986](#)).

Suggested ontology/chemical terms: CHEBI lipopolysaccharide (CHEBI:16412); GO external encapsulating structure / S-layer (GO:0030115).

5. Environmental Information

Infectious agent. *Campylobacter fetus* (NCBI Taxonomy **txid196**), subspecies: - **subsp. fetus (Cff)** — intestinal/genital tract of cattle and sheep; sporadic animal abortion and the main source of human systemic disease ([PMID: 32867004](#)). - **subsp. venerealis (Cfv)** — host-restricted to the bovine genital tract (BGC) ([PMID: 32867004](#)). - **subsp. testudinum (Cft)** — reptile (chelonian: turtle/tortoise) reservoir; emerging cause of human invasive infection ([PMID: 31928704](#); [PMID: 29853499](#); [PMID: 40484837](#)).

Environmental/lifestyle factors. Consumption of **unpasteurized dairy and undercooked/contaminated meat**, occupational/animal contact (livestock, abattoir), and **pet reptile exposure** (for Cft). No toxin, radiation, or pollution etiology applies. The organism is susceptible to zinc-oxide nanoparticles in vitro ([PMID: 30865839](#)) — relevant to control/antimicrobial contexts, not disease causation.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating exposure → clinical manifestation)

1. **Ingestion / mucosal exposure** to *C. fetus* from an animal reservoir (foodborne/zoonotic) **leads to** colonization of the intestinal (or, in cattle, genital) mucosa.
2. Expression of the **paracrystalline S-layer (sapA)**, exported by the **sapCDEF type I secretion system**, **results in** a surface capsule that binds serospecifically to LPS ([PMID: 9851986](#); [PMID: 7885229](#)).
3. The organism **adheres** to mucosal/submucosal surfaces, enhanced by **fibronectin/ECM binding** (S-layer is a major fibronectin-binding site; RGD/integrin-dependent), and **translocates across the intestinal epithelium** (bidirectionally across Caco-2 monolayers, S-layer-independently) **leading to** entry into the bloodstream/submucosa ([PMID: 18388970](#); [PMID: 20600794](#)).
4. In the blood, the **S-layer confers complement/serum resistance** (blocks complement-mediated killing), **resulting in** survival and **sustained high-grade bacteremia** ([PMID: 7885229](#); [PMID: 9851986](#); [PMID: 2318963](#)).
5. Bacteremia **seeds distant sites**, with a hallmark **endovascular tropism** — the branch point — **leading to** mycotic aneurysm, endocarditis, and endovascular graft infection; other

branches seed the **meninges** (meningitis), **joints/bone** (septic arthritis/osteomyelitis), **skin** (cellulitis), **peritoneum** (SBP), and — in pregnancy — the **placenta/fetus** (abortion, stillbirth, neonatal sepsis/meningitis) (PMID: 37877803; PMID: 34849656; PMID: 3523697; PMID: 7246658).

6. **High-frequency antigenic variation** of the S-layer (via invertible DNA element/reciprocal recombination) **results in** immune evasion and **relapse** (~22% in meningitis) (PMID: 7885229; PMID: 41241280).
7. **Host immune status modulates every step**: impaired complement/innate immunity, asplenia, malignancy, and immunosuppression **lead to** higher likelihood of progression to severe/fatal disease (PMID: 2318963; PMID: 29984777; PMID: 34849656).

Inferred vs. demonstrated: Steps 2–4 and 6 are directly demonstrated (in vitro + mouse). The specific molecular basis of endovascular tropism (step 5) is clinically well documented but **mechanistically inferred** (fibronectin/ECM adhesion to damaged/atherosclerotic endothelium is a plausible driver but not proven for *C. fetus* endovascular seeding). CDT/T4SS/fic contributions to human tissue injury are **inferred from genomic presence** rather than demonstrated in human disease.

Category detail

- **Molecular pathways / cellular processes:** complement evasion (S-layer capsule), epithelial translocation, integrin/fibronectin-mediated adhesion (RGD-dependent), CDT-mediated cell-cycle arrest (inferred), T4SS/fic disruption of host-cell processes (inferred). GO suggestions: GO:0030449 (regulation of complement activation — evaded), GO:0007155 (cell adhesion), GO:0009306 (protein secretion), GO:0052572 (response to host immune response).
- **Protein dysfunction:** not host-protein misfolding; rather bacterial S-layer proteins act as a functional immune-evasion armor; outer-membrane vesicles carry the S-layer and are immunoreactive, potentially modulating host response (PMID: 34412928).
- **Immune involvement:** innate immunity (complement, LPS/TLR4 signaling) is central; adaptive immunity limited by antigenic variation. Cell types: intestinal epithelial cells (CL:0000584 enterocyte), vascular endothelial cells (CL:0000115), macrophages/neutrophils (CL:0000235/CL:0000775).
- **Tissue damage mechanisms:** endovascular infection → vessel-wall destruction → mycotic (infected) aneurysm and rupture risk; septic emboli; placental infection → fetal loss.

7. Anatomical Structures Affected

- **Primary sites:** bloodstream (bacteremia). **Endovascular structures** are the signature target: aorta and large arteries (mycotic aneurysm), heart valves/endocardium, vascular grafts/prostheses. UBERON: blood (UBERON:0000178), aorta (UBERON:0000947), endocardium (UBERON:0002165), artery (UBERON:0001637), blood vessel (UBERON:0001981).
 - **Secondary/organ involvement:** meninges/CNS (UBERON:0002360 meninges), joints (UBERON:0001485)/bone, skin & subcutaneous tissue (UBERON:0002097; cellulitis), peritoneum (UBERON:0002358; SBP/ascites), placenta and fetus (UBERON:0001987 placenta), gastrointestinal tract (portal of entry; UBERON:0000160 intestine).
 - **Body systems:** cardiovascular, nervous, musculoskeletal, integumentary, digestive, and reproductive (perinatal).
 - **Tissue/cell level:** intestinal epithelium (translocation); vascular endothelium and ECM/connective tissue (adhesion, damage). Cell Ontology: enterocyte (CL:0000584), endothelial cell (CL:0000115).
 - **Subcellular:** bacterial S-layer/external encapsulating structure (GO:0030115); host extracellular matrix/basement membrane (GO:0005578). Outer-membrane vesicles (bacterial) carry surface antigens.
 - **Lateralization:** not applicable (systemic/bloodborne); aneurysm location varies by patient.
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8. Temporal Development

- **Onset:** predominantly **adult/geriatric** (median ages 68–78); **acute to subacute**. Perinatal disease presents as prolonged maternal febrile illness or fulminant sepsis with transplacental spread ([PMID: 7246658](#)).
- **Progression:** from gut colonization → bacteremia (often within minutes to hours in experimental models) → metastatic seeding over days. Untreated or inadequately treated endovascular disease is **rapidly progressive and can be fatal** ([PMID: 7771897](#)).
- **Course pattern:** acute sepsis; **relapsing-remitting** in a substantial minority (meningitis relapse 22%; relapse notable in HIV/immunocompromised) ([PMID: 41241280](#); [PMID: 9534967](#)).

- **Duration:** with prompt appropriate therapy, treatable/curable; without it, high mortality. Endovascular/CNS infection usually requires prolonged IV antibiotics ± surgery.
- **Critical intervention window:** early appropriate antimicrobial therapy is the key modifiable determinant of survival ([PMID: 34849656](#)).

9. Inheritance and Population

Inheritance: None — acquired infectious disease; no Mendelian pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier-frequency concepts apply.

Epidemiology. - Among *Campylobacter* **bacteremia**, *C. fetus* is a **co-leading species**: 42.6% of 592 cases in a French national 5-year study (vs. *C. jejuni* 42.9%); median age 68 ([PMID: 34849656](#)). An earlier Paris series found *C. fetus* in 53% of 183 episodes ([PMID: 18699745](#)). - *C. fetus* accounts for only ~4% of all *Campylobacter* isolates but has a disproportionately high bacteremia/invasive rate ([PMID: 37877803](#)). - Species other than *C. jejuni*/*C. coli* have higher hospitalization frequency (27.3%); older age, comorbidities, and *C. fetus* infection are associated with bacteremia, and where bacteremia occurs, ~89.5% are hospitalized ([PMID: 41621728](#)). - **Sex ratio:** male predominance (e.g., 78% male in meningitis; 62.45% male in a pooled meta-analysis) ([PMID: 41241280](#); [PMID: 42515013](#)). - **Age distribution:** skewed to older adults; also neonates via vertical transmission. - **Geographic distribution:** worldwide, following animal reservoirs; most published cohorts are from Europe, North America, and East Asia (surveillance bias). Precise incidence/prevalence per 100,000 is **not well quantified** for this species specifically.

10. Diagnostics

Specimen and detection. Because disease is **bacteremic/extra-intestinal**, the key test is **blood culture**, not stool. *C. fetus* is frequently **missed by routine stool Campylobacter protocols**, which incubate at 42 °C — a temperature at which *C. fetus* grows poorly ([PMID: 3175020](#)).

Phenotypic identification / differentiation. | Feature | *C. fetus* | Thermophilic *C. jejuni* |
 |---|---|---| | Growth at 25 °C | Yes | No | | Growth at 42 °C | Poor/No | Yes | | Hippurate
 hydrolysis | **Negative** | Positive | | Nalidixic acid | Resistant (classically) | Sensitive | |
 Cephalothin | Sensitive (classically) | Resistant | | H₂S (TSI) | Negative | — |

Hippurate hydrolysis differentiates *C. jejuni* (positive) from *C. fetus* and other campylobacters (negative) (PMID: 3175020).

Molecular / advanced identification. 16S rRNA sequencing, **MALDI-TOF MS**, multiplex PCR (e.g., **ISCfe1** for subsp. *venerealis*), and **whole-genome sequencing/MLST** for species/subspecies and sequence-type confirmation (e.g., ST80) and resistance-gene prediction (PMID: 22771419; PMID: 32867004; PMID: 40484837).

Imaging. For endovascular disease: CT angiography, transesophageal echocardiography (endocarditis), PET-CT for graft/aneurysm infection (PMID: 7771897).

Genetic testing / omics diagnostics for the patient: Not applicable (no human genetic basis). Pathogen genomics (WGS) is the relevant "omics" diagnostic.

Differential diagnosis. Other causes of Gram-negative bacteremia and endovascular infection (e.g., *Salmonella* mycotic aneurysm), other *Campylobacter*/*Helicobacter* species, culture-negative endocarditis; distinguished by culture, MALDI-TOF, and molecular typing.

11. Outcome / Prognosis

Mortality. 30-day mortality is ~12–15% in adult bacteremia cohorts: 11.7% (French national) (PMID: 34849656), 15% (Paris series) (PMID: 18699745), and 33% (7/21) in a smaller invasive-infection analysis (PMID: 37877803). Meningitis mortality ~5% but relapse 22% (PMID: 41241280). **Perinatal fetal/neonatal mortality ~80%** (PMID: 3523697). Endovascular disease (mycotic aneurysm, prosthetic infection) can be **rapidly fatal** despite surgery and antibiotics (PMID: 7771897).

Prognostic factors. - **Adverse:** cancer (OR 5.1 for death), inappropriate/delayed antibiotics, dyspnea, qSOFA > 2, septic shock, endovascular localization (PMID: 18699745; PMID: 37877803). - **Protective:** appropriate antibiotic therapy (independently associated with 30-day survival; OR 0.47) (PMID: 34849656); favorable outcome in 72% of one series where imipenem was most active (PMID: 17999095).

Morbidity/complications. Mycotic aneurysm (up to 24%), endocarditis, meningitis with neurological sequelae, septic arthritis, cellulitis, SBP, relapse. Systematic reviews report **high antimicrobial resistance, mortality, and relapse risk** for *Campylobacter* bloodstream infection ([PMID: 41820744](#)).

12. Treatment

Pharmacotherapy (first-line). Aminoglycosides (**gentamicin**), carbapenems (**meropenem**, **imipenem**), and **ampicillin** are the mainstays. Imipenem was the most active agent in one series ([PMID: 17999095](#)); perinatal cases were successfully treated with **ampicillin** + **gentamicin** ([PMID: 3523697](#)); post-transplant meningitis was treated with **meropenem** then de-escalated to **high-dose ampicillin** ([PMID: 41241280](#)); postsplenectomy sepsis in an abattoir worker resolved with **meropenem** ([PMID: 29984777](#)).

Resistance considerations (critical). - **Intrinsic cephalosporin resistance** — third-generation cephalosporins should NOT be relied upon (cefotaxime MIC₉₀ 32 mg/L, ceftriaxone 128 mg/L) ([PMID: 22771419](#)). - **Acquired fluoroquinolone resistance** via *gyrA* (Asp→Tyr; T86I) and *parC* mutations; associated with **relapse**, especially in HIV/immunocompromised hosts ([PMID: 9534967](#); [PMID: 22771419](#); [PMID: 40484837](#)). Empiric fluoroquinolone monotherapy is therefore risky. - Tetracycline resistance via *tet(O)* documented in an ST80 strain ([PMID: 40484837](#)).

Surgical/interventional. Endovascular disease (mycotic aneurysm, infected graft/prosthesis, endocarditis) often requires **surgical source control** (aneurysm/graft replacement, valve surgery) in addition to prolonged IV antibiotics ([PMID: 7771897](#); [PMID: 40484837](#)).

Treatment strategy. Prolonged therapy for endovascular/CNS infection; combination therapy (β-lactam + aminoglycoside) commonly used; susceptibility-guided de-escalation. **Timely, appropriate antibiotics are the single most important survival lever** ([PMID: 34849656](#)).

NCIT suggestions: Gentamicin (NCIT:C557), Meropenem (NCIT:C1153), Imipenem (NCIT:C608), Ampicillin (NCIT:C258), Antibiotic Therapy (NCIT:C15832).

Advanced/experimental therapeutics, pharmacogenomics: No gene, cell, RNA, or targeted/immunotherapies apply; no human pharmacogenomic markers established.

13. Prevention

No licensed human vaccine. Human prevention is **host- and exposure-directed**: food safety (avoid unpasteurized dairy and undercooked meat), hygiene around livestock and reptiles, and heightened precaution/early treatment in immunocompromised, asplenic, elderly, and pregnant individuals.

Primary prevention: food-handling and animal-contact hygiene; reptile-exposure counseling (Cft). **Secondary prevention:** prompt blood culture and appropriate therapy in at-risk hosts with fever. **Tertiary prevention:** surgical source control and prolonged antibiotics to prevent endovascular complications/relapse.

Veterinary/One-Health prevention (subsp. *venerealis*, BGC). Control relies on **microbiological testing and culling infected bulls**, with vaccination and antibiotics as adjuncts (PMID: 38441747). Bacterin vaccines are **non-sterilizing**: vaccinated heifers remained culture- and IHC-positive 4 months post-challenge, with Cfv persisting in the reproductive tract/vaginal mucosa (PMID: 42085861). BGC is clinically silent in bulls (persistent preputial carriers), enabling spread absent legislated control (PMID: 26679515); it persists even in vaccinating regions (cultured in 6.6% of bulls; positive farms were non-vaccinators) (PMID: 26412115). Novel multi-epitope (OmpA/FliK) reverse-vaccinology candidates remain in silico only (PMID: 38641595).

14. Other Species / Natural Disease

- **Taxonomy:** *Campylobacter fetus* (NCBI Taxon **txid196**); subspecies *fetus*, *venerealis*, *testudinum*.
- **Natural disease in animals:**
- **Cattle/sheep:** subsp. *fetus* causes **sporadic infectious abortion** and is a main infectious bovine-abortion agent (PMID: 32867004; PMID: 42638448).
- **Cattle (venereal):** subsp. *venerealis* causes **bovine genital campylobacteriosis** — "temporary infertility in female cattle, early embryonic mortality, aberrant oestrus cycles, delayed conception, abortions and poor calving rates" (PMID: 32867004); WOAHL-listed, trade-relevant (PMID: 38133216).
- **Reptiles (chelonians):** subsp. *testudinum* carried by turtles/tortoises (e.g., *Stigmochelys pardalis*) (PMID: 31928704).

- **Zoonotic transmission:** Cff/Cft transmit from animals to humans (foodborne/contact). An invasive human ST80 Cft aortic infection was linked to a **pet soft-shell turtle** — a candidate new host/foodborne vector (PMID: 40484837). Cross-species susceptibility and public-health implications are recognized (PMID: 32998205; PMID: 40638214).
 - **Comparative biology:** virulence machinery (S-layer/*sapA*, T4SS, CDT) is conserved across subspecies; niche restriction (Cfv genital tropism vs. Cff/Cft broader) reflects genomic-island content.
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15. Model Organisms

- **Primary model — mouse.** Adult HA/ICR (outbred) mice **pretreated with ferric chloride (FeCl₃)** recapitulate *C. fetus* bacteremia after oral challenge (PMID: 2318963). This model demonstrated the S-layer's central role: **S-plus strain LD₅₀ was 43.3-fold lower** than its spontaneous S-minus mutant; high-grade bacteremia occurred only with S-plus strains; **anti-S-protein antiserum reduced 30-min bacteremia 51.6-fold**; and **LPS-responsive C3H/HeN mice had 90% mortality vs 40% in LPS-defective C3H/HeJ**, linking host innate immunity to outcome.
 - **Related model — *C. jejuni* mouse.** HA/ICR adult mice model transient bacteremia and gut colonization for enteric campylobacters (contextual comparator) (PMID: 6832823).
 - **In vitro models.** Caco-2 and INT 407 human intestinal epithelial cells for translocation and fibronectin/ECM-mediated adhesion assays (PMID: 20600794; PMID: 18388970); serum-bactericidal assays for complement resistance (PMID: 7885229; PMID: 9851986).
 - **Model characteristics.** The FeCl₃ mouse model **recapitulates bacteremia and S-layer-dependent virulence** but requires iron pretreatment and does not fully reproduce human endovascular tropism or metastatic seeding. Bovine models are used for BGC vaccine/persistence studies (vaccinated-heifer challenge) (PMID: 42085861).
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Key Findings (with evidence)

Finding 1 — Invasive, vascular-tropic bacteremia in vulnerable hosts, with high mortality

C. fetus is a rare but serious pathogen that "mainly affect[s] immunocompromised patients" and causes bacteremia with strong endovascular predilection. In a French invasive-infection study (2000–2021), of 21 bacteremia patients, secondary localizations occurred in 7 (33%), of which **5 were vascular (3 mycotic aneurysms)**, and **7/21 (33%) died within 30 days**; death was associated with dyspnea, qSOFA > 2, and septic shock. The abstract states: "*Secondary localizations were reported for 7 (33%) patients with C. fetus bacteremia, of which 5 exhibited a predilection for vascular infections (including 3 with mycotic aneurysm). Another 7 (33%) patients with C. fetus bacteremia died within 30 days*" (PMID: 37877803). A meningitis review of 37 cases found 78% male, median age 49, relapse 22%, mortality 5% (PMID: 41241280).

Finding 2 — The S-layer (sapA) is the central virulence mechanism

The paracrystalline S-layer confers **complement/serum resistance** and undergoes **high-frequency antigenic variation** to facilitate persistent colonization: "*Campylobacter fetus utilizes paracrystalline surface (S-) layer proteins that confer complement resistance and that undergo antigenic variation to facilitate persistent mucosal colonization in ungulates*" (PMID: 7885229). The S-layer is exported by a **type I secretion system (sapCDEF)** without an N-terminal signal sequence, and "*the virulence of Campylobacter fetus... is mediated in part by the presence of a paracrystalline surface layer (S-layer) that confers serum resistance*" (PMID: 9851986). *sapA* disruption abolishes the S-layer and reduces serum survival; revertants restore serum resistance and cause ~10-fold more bacteremia in mice (PMID: 7885229).

Finding 3 — Subspecies, niches, and additional virulence factors

Cff inhabits the cattle/sheep intestinal/genital tract and causes human systemic disease; *Cfv* is host-restricted to the bovine genital tract: "*Cfv causes syndrome of temporary infertility in female cattle, early embryonic mortality, aberrant oestrus cycles, delayed conception, abortions and poor calving rates*" (PMID: 32867004). *C. fetus* carries conserved **T4SS** and **fic-domain** genes on genomic islands/plasmids that "*may disrupt host cell processes*" (PMID: 27049518), and invasive strains carry *ciaB*, *cdtABC*, T4SS, and *sapA* (PMID: 40484837). The organism "*was found to translocate equally well in both apical-to-basolateral and basolateral-to-apical directions*" across Caco-2 epithelium, explaining gut-to-blood dissemination (PMID: 20600794).

Finding 4 — Perinatal disease with high fetal mortality

Maternal bacteremia originating from the bowel produces feto-placental involvement: *"Eighteen of 20 pregnancies... ended prematurely at 13-32 weeks of gestation. All of the mothers survived, but fetal/neonatal mortality was 80%"* (PMID: 3523697). Transplacental spread *"may result in abortion, stillbirth, or early neonatal meningitis"* (PMID: 7246658). Perinatal cases were successfully treated with ampicillin + gentamicin.

Finding 5 — Treatment and resistance

Treatment relies on aminoglycosides/carbapenems/ampicillin. Quinolone resistance arises via a *gyrA* mutation: *"a G-to-T change that led to an Asp-to-Tyr amino acid substitution at a critical residue frequently associated with quinolone resistance"*, linked to relapse in HIV patients (PMID: 9534967). In a Taiwan bacteremia series, *"the majority of the isolates were resistant to third-generation cephalosporins and quinolones"*, with *parC* mutations (PMID: 22771419).

Finding 6 — Diagnosis by blood culture with characteristic phenotype

C. fetus is optimally recovered from **blood culture** and often missed by 42 °C stool protocols; it is **hippurate-negative** (differentiating it from *C. jejuni*): *"Hippurate hydrolysis was used to differentiate C. jejuni (positive) from all the other Campylobacter spp (negative)"* (PMID: 3175020). Species/subspecies confirmation uses 16S/multiplex PCR/MALDI-TOF/WGS (PMID: 22771419).

Finding 7 — Mouse model proves S-layer centrality

In the FeCl₃-pretreated HA/ICR mouse model, *"the LD50 for S-plus strain 84-32 was 43.3 times lower than its spontaneous S-minus mutant 84-54"*, and *"these findings in a mouse model point toward the central role of the S-protein in the pathogenesis of C. fetus infection"* (PMID: 2318963). Anti-S antiserum reduced bacteremia 51.6-fold, and host LPS-responsiveness modulated mortality (C3H/HeN 90% vs C3H/HeJ 40%).

Finding 8 — Emerging reptile-associated subsp. *testudinum*

Cft is carried by chelonians — surveys detected *"one for C. fetus subsp. testudinum (Stigmochelys pardalis)"* (PMID: 31928704) — has been isolated from human ascites in chronic kidney disease (PMID: 29853499), and an invasive ST80 strain (nalidixic acid/ciprofloxacin/tetracycline resistant; *gyrA* T86I, *tet(O)*; carrying *ciaB*, *cdtABC*, T4SS, *sapA*) caused thoracoabdominal aortic infection, suspected source a pet turtle (PMID: 40484837).

Finding 9 — Host-specific prevention; non-sterilizing bovine vaccines

BGC control = test-and-cull + adjunct vaccination: *"The control of both diseases relies on microbiological testing and culling infected bulls"* (PMID: 38441747). Vaccination is non-sterilizing — *"vaccinated animals also remained culture- and IHC-positive at four months post-infection, suggesting that vaccination did not induce sterilizing immunity and did not prevent persistence of Cfv within the reproductive tract"* (PMID: 42085861); BGC persists even where vaccination is recommended (PMID: 26412115). No licensed human vaccine exists.

Finding 10 — Epidemiology and prognosis

C. fetus is a co-leading cause of *Campylobacter* bacteremia — *"Campylobacter jejuni and Campylobacter fetus were the most commonly identified species (in 42.9% and 42.6%, respectively). The patients were elderly (median age 68 years)";* appropriate antibiotics were protective (OR 0.47) (PMID: 34849656). Compared with other species, *"patients with C. fetus bacteremia were older (mean age, 69.5 years vs. 55.6 years; P = .001) and were more likely to have cellulitis (19% vs. 7%; P = .03), endovascular infection (13% vs. 1%; P = .007)"* (PMID: 18699745); mycotic aortic aneurysm reached 24% in one series (PMID: 17999095).

Finding 11 — Integrated S-layer × host-immunity model

Convergent in-vitro, mouse, and clinical evidence establishes upstream S-layer serum resistance → bacteremia → endovascular/CNS/placental seeding → antigenic-variation-driven relapse, modulated by host immunity. The mouse model *"point[s] toward the central role of the S-protein in the pathogenesis of C. fetus infection"* (PMID: 2318963), while the clinical lever is that *"an appropriate antibiotic treatment was independently associated with 30-day survival"* (PMID: 34849656).

Mechanistic Model / Interpretation

```

Animal reservoir (cattle/sheep; reptiles for Cft)
  |  foodborne / contact (zoonotic)
  v
Gut colonization --(fibronectin/ECM, RGD-integrin adhesion; <a href="https://pubmed.ncbi.nlm.nih.gov/37877803/" rel="noopener noreferrer" title="Gut colonization in cattle/sheep; reptiles for Cft")
  |
  v  epithelial translocation (Caco-2, S-layer-independent; <a href="https://pubmed.ncbi.nlm.nih.gov/17999095, 34849656/" rel="noopener noreferrer" title="Epithelial translocation in Caco-2, S-layer-independent")
BLOODSTREAM ENTRY
  |
  v  S-LAYER (sapA) -> complement/serum resistance  <-- HOST IMMUNITY
  |  (<a href="https://pubmed.ncbi.nlm.nih.gov/7885229/" rel="noopener noreferrer" title="S-layer (sapA) -> complement/serum resistance")
  v  malignancy, HIV, age)
SUSTAINED BACTEREMIA -----+
  |
  +--> ENDOVASCULAR (hallmark): mycotic aneurysm,      | antigenic variation
  |      endocarditis, graft infection (<a href="https://pubmed.ncbi.nlm.nih.gov/37877803/" rel="noopener noreferrer" title="Endocarditis, graft infection")
  |      17999095, 34849656)      | -> immune evasion -> RELAPSE
  +--> MENINGES: meningitis (<a href="https://pubmed.ncbi.nlm.nih.gov/41241280/" rel="noopener noreferrer" title="Meninges: meningitis")
  +--> JOINTS/BONE, SKIN (cellulitis), PERITONEUM (SBP) |
  +--> PLACENTA/FETUS: abortion, stillbirth, neonatal |
  |      sepsis/meningitis (<a href="https://pubmed.ncbi.nlm.nih.gov/3523697/" rel="noopener noreferrer" title="Placenta/fetus: abortion, stillbirth, neonatal")
  |
  v  MODIFIABLE LEVER: timely appropriate antibiotics (OR 0.47; <a href="https://pubmed.ncbi.nlm.nih.gov/3523697/" rel="noopener noreferrer" title="Modifiable lever: timely appropriate antibiotics")
OUTCOME: ~12-15% 30-day mortality (adults); ~80% fetal/neonatal mortality (perinatal)

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Upstream vs downstream. The **S-layer/serum-resistance axis** is **upstream** and necessary for bloodstream persistence; **metastatic seeding, tissue destruction, and relapse are downstream** consequences. **Host immune competence is a parallel upstream modulator** that gates progression at every step. The **actionable node** is prompt, susceptibility-appropriate antibiotic therapy (avoiding cephalosporins; cautious with fluoroquinolones), plus surgical source control for endovascular disease.

Evidence Base

PMID	Study type	Contribution
37877803	Retrospective clinical (France)	Vascular tropism & 33% 30-day mortality in invasive disease
34849656	National bacteremia cohort	<i>C. fetus</i> co-leads bacteremia; 11.7% mortality; appropriate antibiotics protective
18699745	Clinical cohort	Older age, endovascular tropism (13% vs 1%), 15% mortality
17999095	Clinical series	Mycotic aneurysm 24%, cellulitis 19%, imipenem most active
41241280	Case + literature review	Meningitis: immunocompromised hosts, relapse 22%
7885229	In vitro/molecular	S-layer confers complement resistance & antigenic variation
9851986	Molecular	S-layer secreted by type I secretion system; serum resistance
2318963	Mouse model	S-protein central to virulence; host LPS-responsiveness modulates mortality
20600794	In vitro (Caco-2)	Bidirectional epithelial translocation (gut→blood)
18388970	In vitro	Fibronectin/ECM adhesion via S-layer, RGD-integrin dependent
27049518	Genomics	T4SS & fic-domain virulence factors
40484837	Genomics/case (ST80 Cft)	Reptile-associated invasive aortic infection; resistance & virulence genes
32867004	Veterinary genomics	Cfv host-restricted BGC; ISCfe1 typing
3523697	Case series/review	Perinatal: 80% fetal/neonatal mortality
7246658	Case report/review	Transplacental spread → abortion/stillbirth/neonatal meningitis
9534967	Molecular/clinical	gyrA-mediated quinolone resistance & relapse in HIV
22771419	Bacteremia/AST series	Cephalosporin & quinolone resistance; parC mutations

PMID	Study type	Contribution
3175020	Microbiology	Hippurate-negative differentiation from <i>C. jejuni</i>
31928704	Veterinary survey	Cft carriage in chelonians
29853499	Genome/case	Cft from human ascites (CKD)
42085861	Bovine challenge	Non-sterilizing BGC vaccination
38441747	Veterinary program	Test-and-cull BGC control
29984777	Case report	Asplenia/rituximab/occupational risk; meropenem cure
41621728	Meta-analysis	Non-jejuni/coli higher hospitalization; <i>C. fetus</i> ↔bacteremia
41820744	Systematic review	High AMR, mortality, relapse in <i>Campylobacter</i> BSI

Evidence source types: human clinical (cohorts, case series/reviews), veterinary/animal-reservoir studies, in-vitro molecular/cell-culture, and mouse-model experiments. No human computational/GWAS evidence applies (no heritable basis).

Limitations and Knowledge Gaps

1. **No human genetic architecture.** Sections 4/9 on causal genes, inheritance, penetrance, and carrier frequency are **not applicable**; all "genetic" content concerns bacterial virulence/resistance genes.
2. **Retrospective, surveillance-biased epidemiology.** Cohorts are predominantly European/North American/East Asian; global incidence/prevalence per 100,000 is not well quantified, and *C. fetus*-specific (vs. genus-level) rates are sparse.
3. **Mechanism of endovascular tropism is inferred.** Fibronectin/ECM adhesion is a plausible driver, but the molecular basis for *C. fetus*'s specific affinity for damaged/atherosclerotic vessels is unproven.
4. **CDT/T4SS/fic roles in human disease are genomic inferences**, not demonstrated in human tissue.
5. **Small perinatal and meningitis series** yield wide uncertainty around mortality/relapse estimates.

6. **No human vaccine and limited pharmacogenomics;** treatment evidence is largely observational (no RCTs).
7. **Model limitations:** the FeCl₃ mouse model requires iron pretreatment and does not reproduce endovascular seeding; no widely used organoid or humanized model exists.

Proposed Follow-up Experiments / Actions

1. **Quantify population burden:** systematic estimate of *C. fetus*-specific incidence/prevalence and case-fatality by age/immune status using linked national bacteremia registries and GBD-style modeling.
2. **Dissect endovascular tropism:** endothelial/aortic-explant adhesion-invasion assays ($\pm$ fibronectin, $\pm$ RGD blockade, $\pm$ S-layer) and a vascular-injury animal model to test whether ECM exposure drives mycotic-aneurysm seeding.
3. **Resistance surveillance:** prospective WGS-based tracking of *gyrA/parC/tet(O)* and MIC trends to inform empiric regimens and stewardship, given cephalosporin intrinsic resistance and rising fluoroquinolone resistance.
4. **Antigenic-variation & relapse:** longitudinal genomic tracking of *sapA* inversion/recombination in relapsing (especially HIV/meningitis) patients to link S-layer switching to clinical relapse.
5. **Host-factor immunology:** evaluate complement pathway, splenic function, and innate-immune status as predictors of progression to invasive disease; assess whether biomarkers stratify risk.
6. **One-Health / reptile reservoir:** characterize prevalence and transmission of subsp. *testudinum* from pet/food reptiles and define public-health messaging for immunocompromised owners.
7. **Vaccine science (veterinary and human):** advance multi-epitope (OmpA/FliK) and S-layer/OMV-based candidates from in-silico to in-vivo testing, aiming for sterilizing bovine immunity and exploring feasibility of human high-risk-group vaccination.

Report compiled from 11 confirmed findings and 45 reviewed papers across 5 investigation iterations. Ontology suggestions (HPO, GO, CL, UBERON, CHEBI, NCIT, NCBI Taxon) are provided inline per section for knowledge-base ingestion.

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